

Asset-Liability Management of Foreign Exchange Reserves: A Factor Model Approach

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December 2017

Forex reserves in EM: When are they needed?

- ▶ Eichengreen, Hausmann, and Panizza (2005): ‘original sin’ as the inability of a country to borrow abroad in its own currency

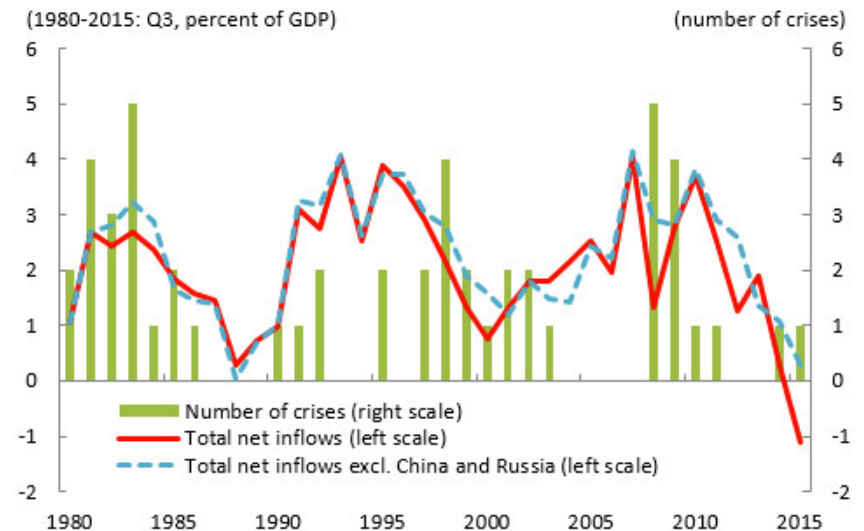


Economist.com

Net capital flows and crises

Declining net capital flows to emerging markets during 2010-15 didn't bring a higher incidence of debt crises as similar slowdowns in the past.

(1980-2015: Q3, percent of GDP)



Source: IMF April 2016 *World Economic Outlook*.

Problem?

- ▶ Reserves in EM countries are needed when countries are in:
 - ▶ Liquidity/Banking crisis; Financial contagion; Commodity swings
- ▶ Assets that are valuable in these scenarios have low yields in normal periods:
 - ▶ Gold
 - ▶ Government bonds of developed countries
 - ▶ Put derivatives
- ▶ $\Rightarrow$ Holding reserves is costly $\approx 1\%$ GDP (Rodrik, 2006):
 - ▶ 'Yield give up' example: Indian 10Y Bond (7.1%) versus US 10Y (2.3%).

Contribution of this Paper

- ▶ Normative asset-liability management model of Forex reserves:
 - ▶ Optimal factor exposure to systemic risks:
 - ▶ Yield-give up;
 - ▶ Hedge: i.e. liquidity risk in foreign currency.
 - ▶ Replicating asset portfolio (constraints);
 - ▶ Capital preservation;
 - ▶ Evaluation of strategies.
- ▶ Practical implementation of the model to the Chilean case:
 - ▶ A small-open economy with significant exposure to copper prices.

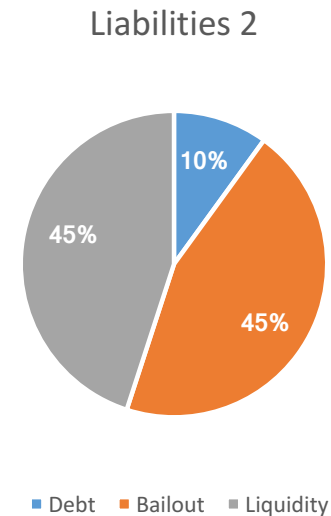
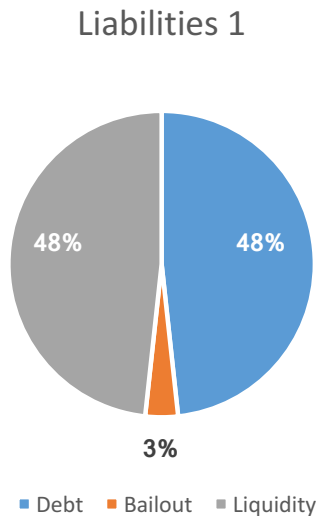
Model

1. Measuring liabilities:

- ▶ Observables: opportunity cost (i.e. cost of government debt).
- ▶ Unobservables:
 - liquidity provision (swaption on spread in foreign-currency).
 - bail-out cost of financial sector (put option on banks' assets).

Methodology: Measuring explicit and contingent liabilities

- ▶ Estimated Total Central Bank liabilities composition:



Contribution of this Paper

- ▶ The model:
 - 2. Identifying global systematic factors;

Risk Factors

- ▶ Global Fama-French 3 Factor: $\text{MRP} + \text{SMB} + \text{HML}$
- ▶ Global Fama-French 5 Factor: $\text{MRP} + \text{SMB} + \text{HML} + \text{RMW} + \text{CMA}$
- ▶ Chen, Ross and Roll (1986): $\text{MRP} + \text{InfSurp} + \text{IndProduSurp} + \text{TP} + \text{ExpInfl}$
- ▶ Macro Model I: $\text{Economic} + \text{Inflation} + \text{EM Curr}$
- ▶ Macro Model II: $\text{Economic} + \text{Credit} + \text{EM MRP}$
- ▶ Macro Model III: $\text{EM Curr} + \text{EM MRP}$
- ▶ Macro Model IV: $\text{Real Rates} + \text{TP}$

* RMW: (Robust Minus Weak) is the average return on the two robust operating profitability portfolios minus the average return on the two weak operating profitability portfolios;

** CMA (Conservative Minus Aggressive) is the average return on the two conservative investment portfolios minus the average return on the two aggressive investment portfolios,

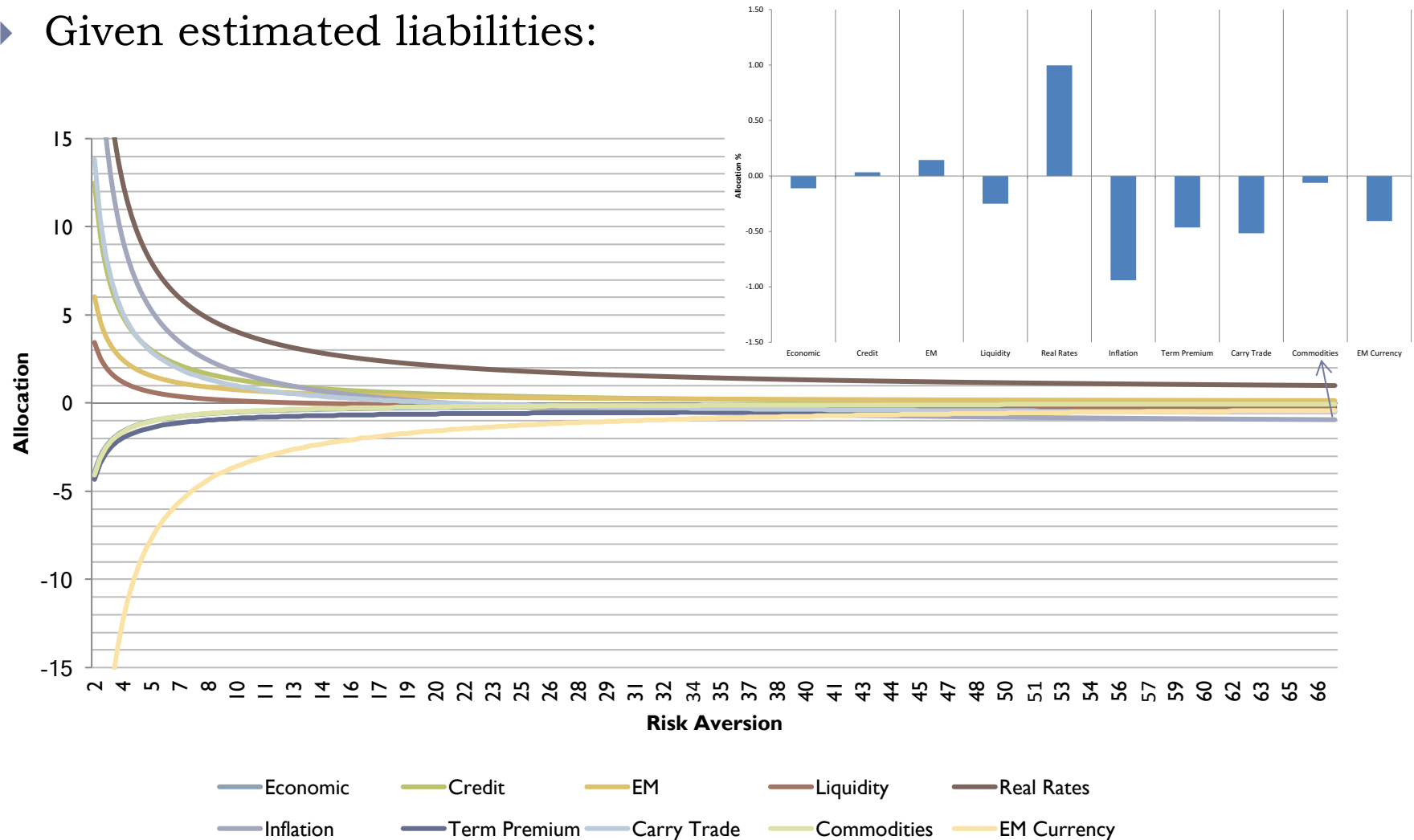
Contribution of this Paper

- ▶ The model:

3. Optimal exposure to global systematic factors;

Optimal Factor Allocation

► Given estimated liabilities:



Contribution of this Paper

- ▶ The model:

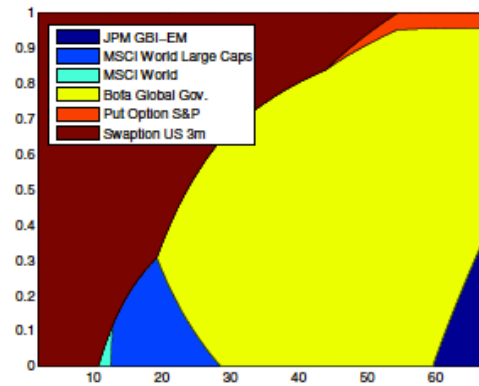
4. Investable assets that “better” replicate the exposure to global factors;

Factor Replication

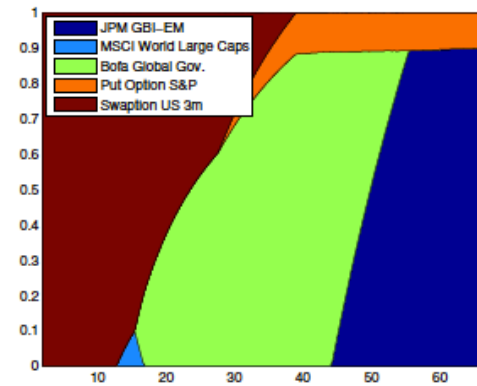
	Constant	Economic	Real Rates	Credit	EM	Term Premium	Size	Carry Trade	EM Currency	Commodities	R2
Barclays World Inflation Linked Bonds TR	0.0%	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
BofA Merrill Lynch Australian Govt	0.4%	0.02	0.08	-0.01	-0.01	0.18	0.22	0.04	-0.09	-0.01	0.52
BofA Merrill Lynch Canada Government Ind	0.3%	0.01	0.08	-0.01	0.02	0.09	0.28	0.02	-0.01	0.00	0.67
BofA Merrill Lynch Euro Government Index	0.3%	0.00	0.12	0.00	0.01	0.17	0.20	-0.02	0.09	-0.05	0.47
BofA Merrill Lynch Global Corporate Inde	0.1%	0.04	0.74	0.15	-0.01	0.54	0.04	-0.02	0.06	0.02	0.86
BofA Merrill Lynch Global Government Ind	0.0%	0.00	1.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	1.00
BofA Merrill Lynch Global High Yield Ind	0.2%	-0.01	0.38	0.95	0.00	0.22	0.06	-0.01	0.06	-0.01	0.97
BofA Merrill Lynch Japan Government Inde	0.1%	-0.03	0.10	0.01	0.01	0.19	0.04	0.00	0.08	-0.01	0.12
BofA Merrill Lynch New Zealand Govt	0.4%	0.02	0.12	0.02	-0.01	0.14	0.15	0.01	-0.09	-0.01	0.41
BofA Merrill Lynch Switzerland Governmen	0.2%	-0.02	0.06	0.01	0.03	0.05	0.21	0.01	0.02	-0.03	0.41
BofA Merrill Lynch UK Gilt Index	0.3%	0.03	0.10	-0.01	0.03	0.06	0.39	-0.03	-0.02	-0.04	0.63
BofA Merrill Lynch US Treasury Index	0.2%	-0.01	0.14	-0.08	0.00	0.10	0.34	0.00	0.01	0.00	0.90
CLPUSD Spot Exchange Rate - Price of 100	-0.3%	-0.04	0.51	0.26	0.13	0.60	-0.11	-0.01	0.21	-0.01	0.40
China Money Market	0.4%	-0.01	0.03	-0.01	0.00	0.00	-0.02	-0.02	-0.01	0.00	0.02
Copper	0.7%	0.25	0.64	0.07	0.27	0.61	-0.76	-0.18	0.39	0.63	0.47
J.P. Morgan ELMi Plus Chile	0.2%	-0.04	0.58	0.23	0.16	0.59	-0.16	-0.03	0.22	-0.02	0.38
J.P. Morgan EMBI Global Total Return Ind	0.7%	0.19	-0.07	0.37	0.25	0.27	0.39	-0.19	0.13	0.01	0.55
J.P. Morgan EMBI Plus Mexico	0.6%	0.16	0.02	0.24	0.14	0.27	0.52	-0.14	0.06	-0.01	0.57
JPM GBI Global Total Return Index Level	0.0%	0.00	0.99	-0.01	0.00	1.00	0.01	0.02	-0.03	-0.01	0.98
Korea Money Market	0.0%	0.33	0.75	0.07	0.14	0.34	-0.23	0.04	0.04	-0.06	0.30
MSCI AC Asia Pacific Index	0.1%	0.55	0.47	0.38	0.35	0.39	-0.19	-0.10	-0.07	0.11	0.60
MSCI ACWI Index	0.4%	0.61	0.24	0.53	0.06	0.32	-0.20	-0.35	0.13	0.07	0.73
MSCI Chile Index	0.6%	0.24	0.25	0.69	0.70	0.84	-0.26	-0.26	0.31	0.00	0.59
MSCI Emerging Markets Index	0.4%	0.62	0.22	0.52	0.99	0.32	-0.20	-0.34	0.12	0.07	0.88
MSCI World Index	0.4%	0.62	0.22	0.52	-0.01	0.32	-0.20	-0.34	0.12	0.07	0.71
MSCI World Large Cap Index	0.4%	0.61	0.23	0.53	0.00	0.33	-0.20	-0.38	0.12	0.07	0.70
MSCI World Small Cap Index	0.4%	0.61	0.23	0.53	0.00	0.33	-0.20	0.62	0.12	0.07	0.77
Nikkei 225	0.1%	0.58	0.26	0.31	0.05	0.10	-0.04	0.14	-0.31	0.16	0.29
Oil	1.1%	-0.11	-0.16	-0.14	0.20	-0.99	-0.34	0.18	-0.22	1.53	0.58
XAUUSD Spot Exchange Rate - Price of 1 X	0.4%	-0.04	0.80	-0.20	0.20	0.60	0.09	0.15	-0.14	0.36	0.34

Optimal Replicating Portfolio

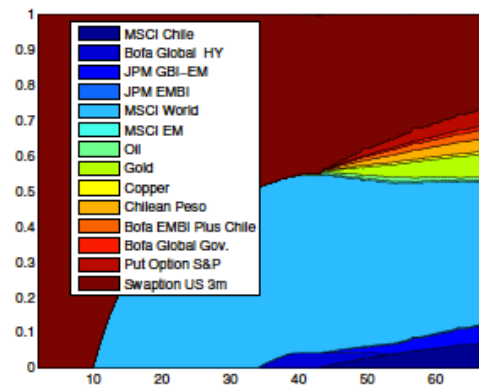
► All assets:



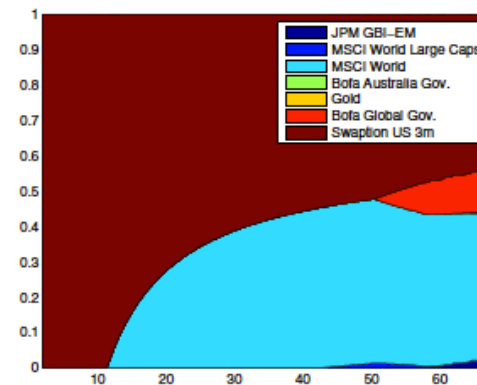
(a) Liabilities I



(b) Liabilities II



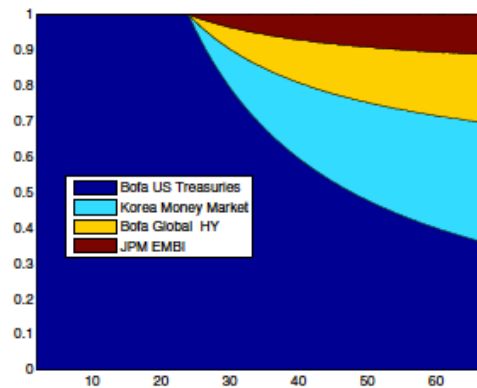
(c) Liabilities III



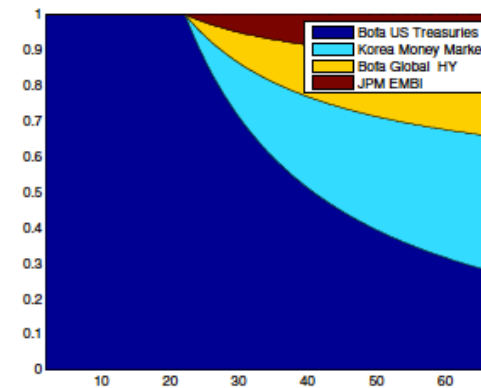
(d) Liabilities IV

Optimal Replicating Portfolio

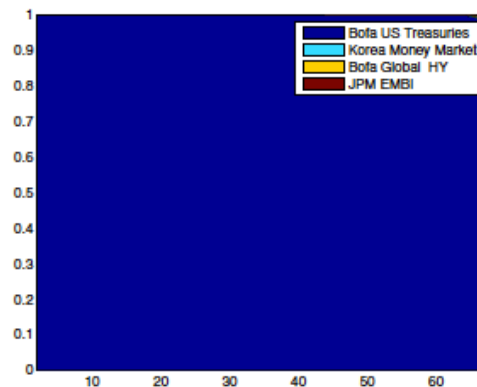
► Fixed income only:



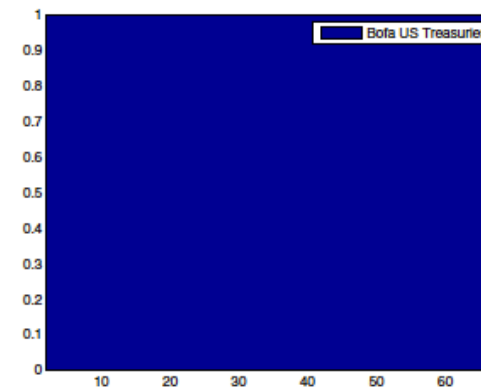
(a) Liabilities I



(b) Liabilities II



(c) Liabilities III



(d) Liabilities IV

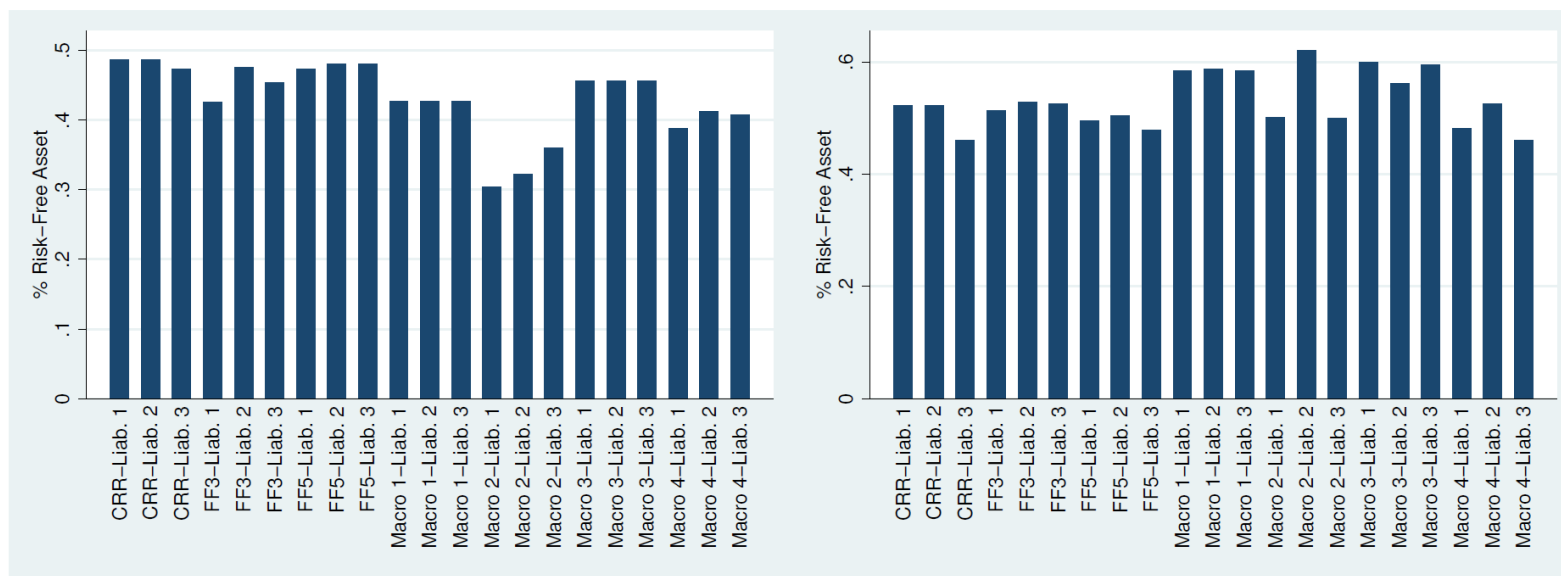
Contribution of this Paper

- ▶ The model:

5. Capital preservation: Protective put strategy.

Protective put strategy

- ▶ Following Merton (1981):
 - ▶ Higher the risk of the portfolio → Higher the allocation in the risk-free asset.



(a) % Invested in the Risk-Free Asset - Fixed Income Strategies

(b) % Invested in the Risk-Free Asset - All Assets

*Risk-free asset: 3 month Treasury Bills.

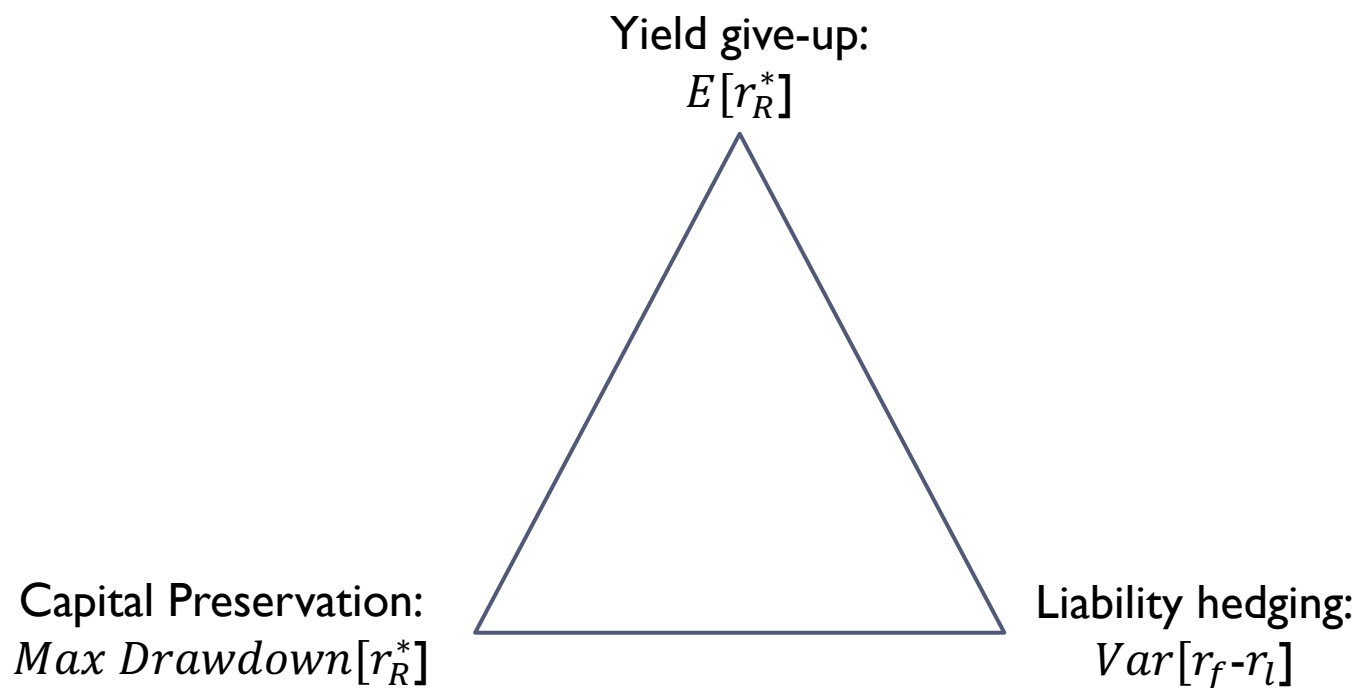
Contribution of this Paper

- ▶ The model:

6. Comparing investment strategies: i) Hedging liabilities; ii) Potential capital losses; iii) Yield-give up.

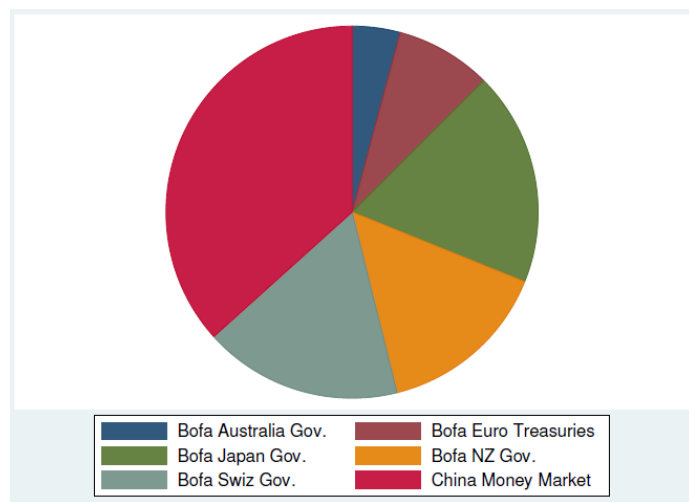
Ex-ante Evaluation

- ▶ Depending on:
 - ▶ Relevant systematic factors;
 - ▶ Measurement of contingent liabilities;
 - ▶ Investable assets;
- ▶ How Central Banks can decide their optimal strategy?

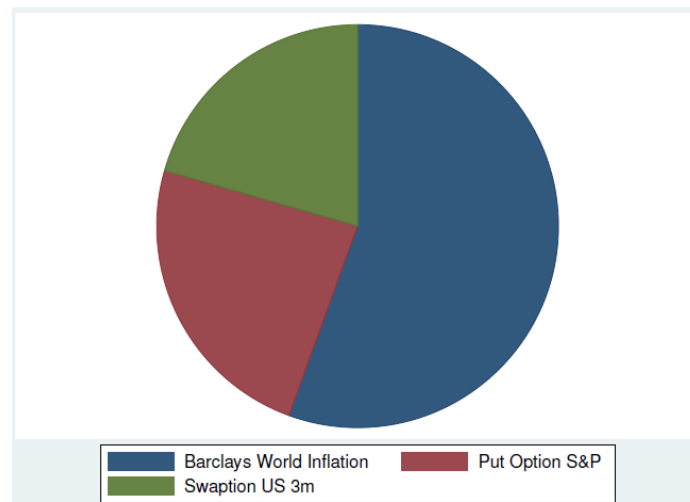


Ex-ante Evaluation

- ▶ Central Bank preferences can be incorporated:
 - ▶ Exogenous weights;
 - ▶ Equal weights; PCA-weighting; z-scores weights, (Chincarini, 2006).



Fixed Income Spectrum



Non-restricted Spectrum

Final Remarks

- ▶ The investment problem of Central Banks' reserves includes:
 - ▶ Observable and unobservable liabilities that have to be hedged;
 - ▶ Risk-return trade-off subject to a usually strong capital preservation motive.
 - ▶ Institutional restrictions limit the investable asset-set.
- ▶ My normative model aboard these issues, maintains quantitative tractability, and offers an internally consistent self-evaluation methodology.
- ▶ Specifically, in the case of Chile, I find that:
 - ▶ There are potential gains of including derivatives in the foreign exchange reserves portfolio: Put options on the S&P500 or Swaptions on USD interest rate swaps.

Appendix

Modern theory on Forex reserves

- ▶ Foreign exchange intervention that are part of a development strategy (i.e. Rajan & Subramanian (2011): “dutch disease” prevention).
- ▶ Correcting occasional short-term misalignments, Daude et al. (2016).
- ▶ Precautionary motive. Hedging against “sudden stops”, Aizenman & Lee (2007).

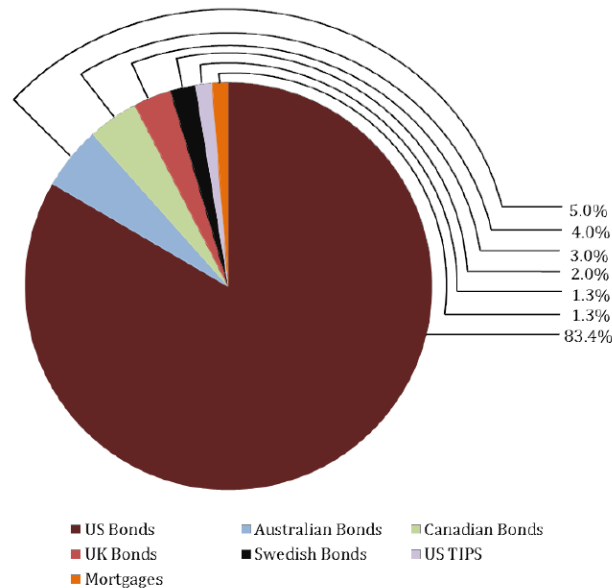
How much reserves?

- ▶ 1980s: three months' worth of imports rule-of-thumb.
- ▶ Guidotti–Greenspan (1999): short-term external debt.
- ▶ Caballero & Panageas (2008): 10% GDP in reserves.
- ▶ IMF (2011): multivariate linear function on monetary base, short-term external debt, other relevant contingent liabilities and imports.

Appendix

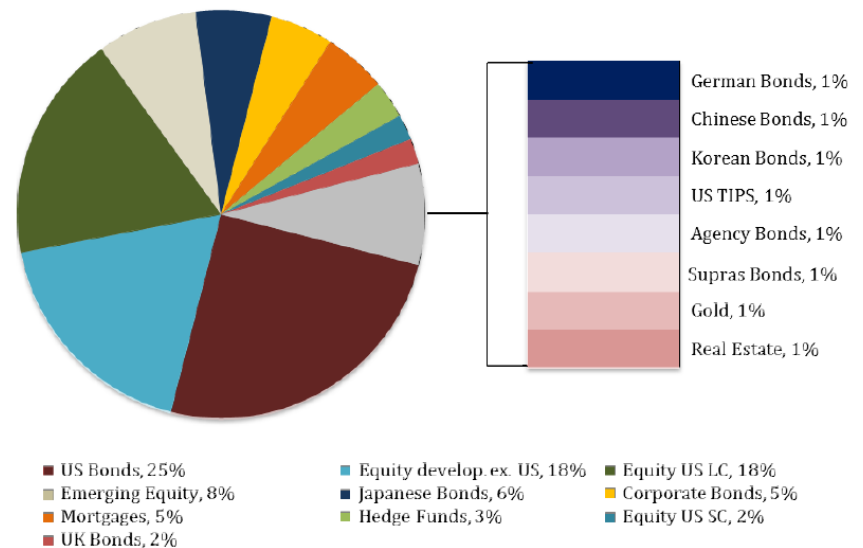
- García, Gomez & Vela (2015). An Asset Allocation Framework with Tranches for Foreign Reserves. Working Paper, Central Bank of Colombia.

Figure 1: Asset allocation of the Safety Tranche



Source: Authors' calculations

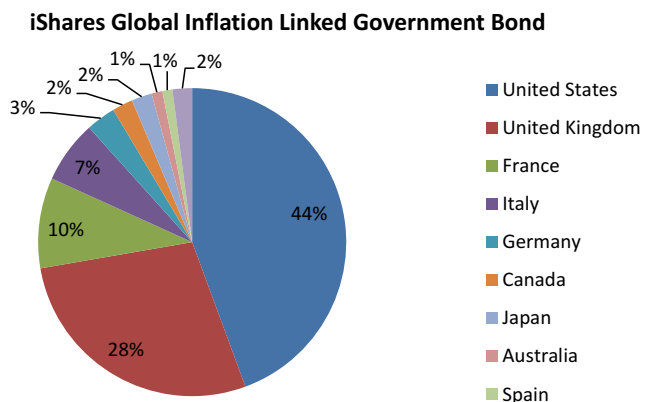
Figure 2: Wealth Tranche asset allocation



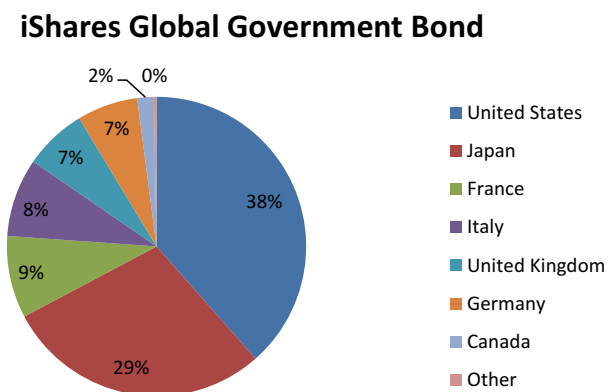
Source: Authors' calculations

Appendix

Dur.12,43



Dur.8,12



Appendix

- ▶ Zhang, Zhang & Zhang (2013). Strategic Asset Allocation for China's Foreign Reserves: A Copula Approach. China & World Economy / 1–21, Vol. 21, No. 6.

Table 6. Asset Allocation Based on Copula with Conditional Value-at-risk (CVaR) Minimization, 2006–2012

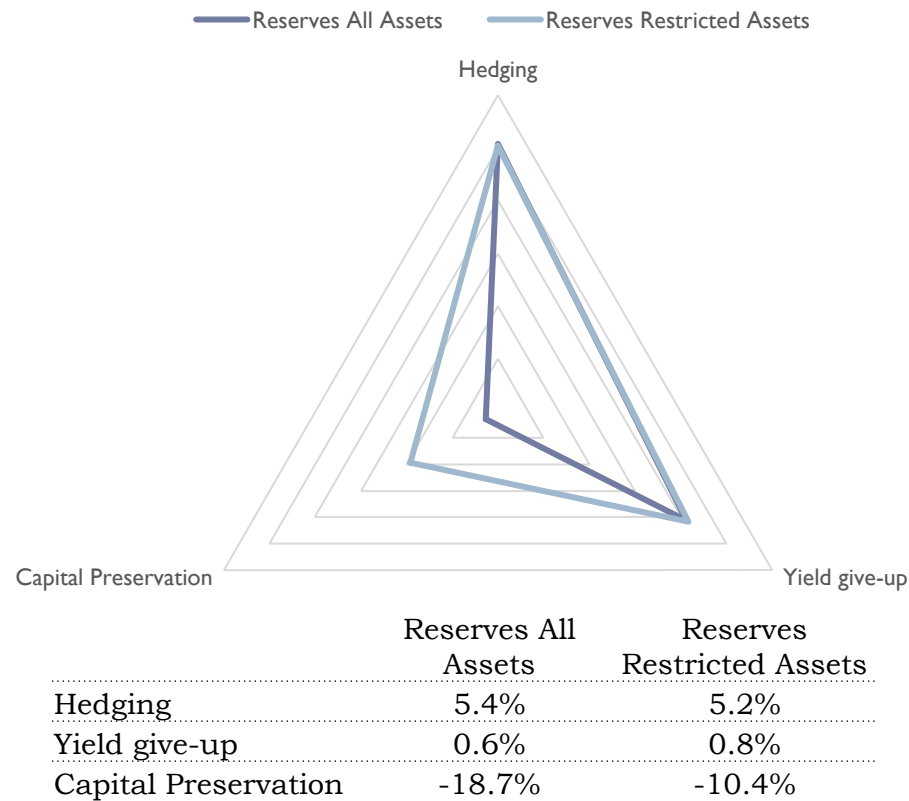
		2006	2007	2008	2009	2010	2011	2012
Short-term	Treasury bond (%)	0.86	0.71	5.71	10.71	15.70	20.70	25.69
	Deposit (%)	7.21	12.20	17.20	22.20	27.20	32.20	37.20
	Treasury bond (%)	35.88	30.88	25.88	20.88	15.89	10.89	10.03
Long-term	Agency debt (%)	30.38	25.38	29.61	30.75	29.57	30.23	26.38
	Corporate bond (%)	0.30	0.00	0.77	0.00	1.19	0.00	0.00
	Equity (%)	0.38	5.37	0.37	0.00	0.00	0.53	0.24
	Euro (%)	25.00	25.45	20.45	15.45	10.45	5.45	0.45
	Mean	−0.00028	0.000544	0.001285	0.001526	0.000317	0.000607	0.000398
	CVaR	−0.15193	−0.17216	−0.3135	−0.17405	−0.19208	−0.09707	−0.04335

Table 7. Asset Allocation in Copula Model with Disappointment Avoidance Maximization, 2006–2012

		2006	2007	2008	2009	2010	2011	2012
Short-term	Treasury bond (%)	1.30	0.04	4.97	0.15	0.80	0.06	0.08
	Deposit (%)	7.13	11.96	16.92	17.64	19.46	14.55	14.16
	Treasury bond (%)	35.91	30.93	26.49	31.39	36.31	41.29	46.26
Long-term	Agency debt (%)	30.22	25.25	21.04	16.13	12.20	12.98	8.33
	Corporate bond (%)	0.32	0.01	0.72	0.02	0.11	0.00	0.01
	Equity (%)	0.15	5.14	2.33	2.17	3.61	8.59	13.58
	Euro (%)	24.97	26.67	27.53	32.51	27.51	22.53	17.58
	Mean	−0.0003	0.0005	0.0017	0.0025	0.0005	0.002	0.0022
	CVaR	−0.1515	−0.1737	−0.3983	−0.2864	−0.4601	−0.3133	−0.2314

Strategic evaluation

- Finally, we can perform an evaluation of the decision to impose a restriction on the investable assets:



* i) The hedging metric is calculated as the historical standard deviation of the excess of return of the portfolio of reserves over the liabilities; ii) The yield give-up is equal to the product between the estimated betas and the risk-premium (in-sample); iii) The capital preservation metric is equal to the maximum drawdown of the reserves portfolio.

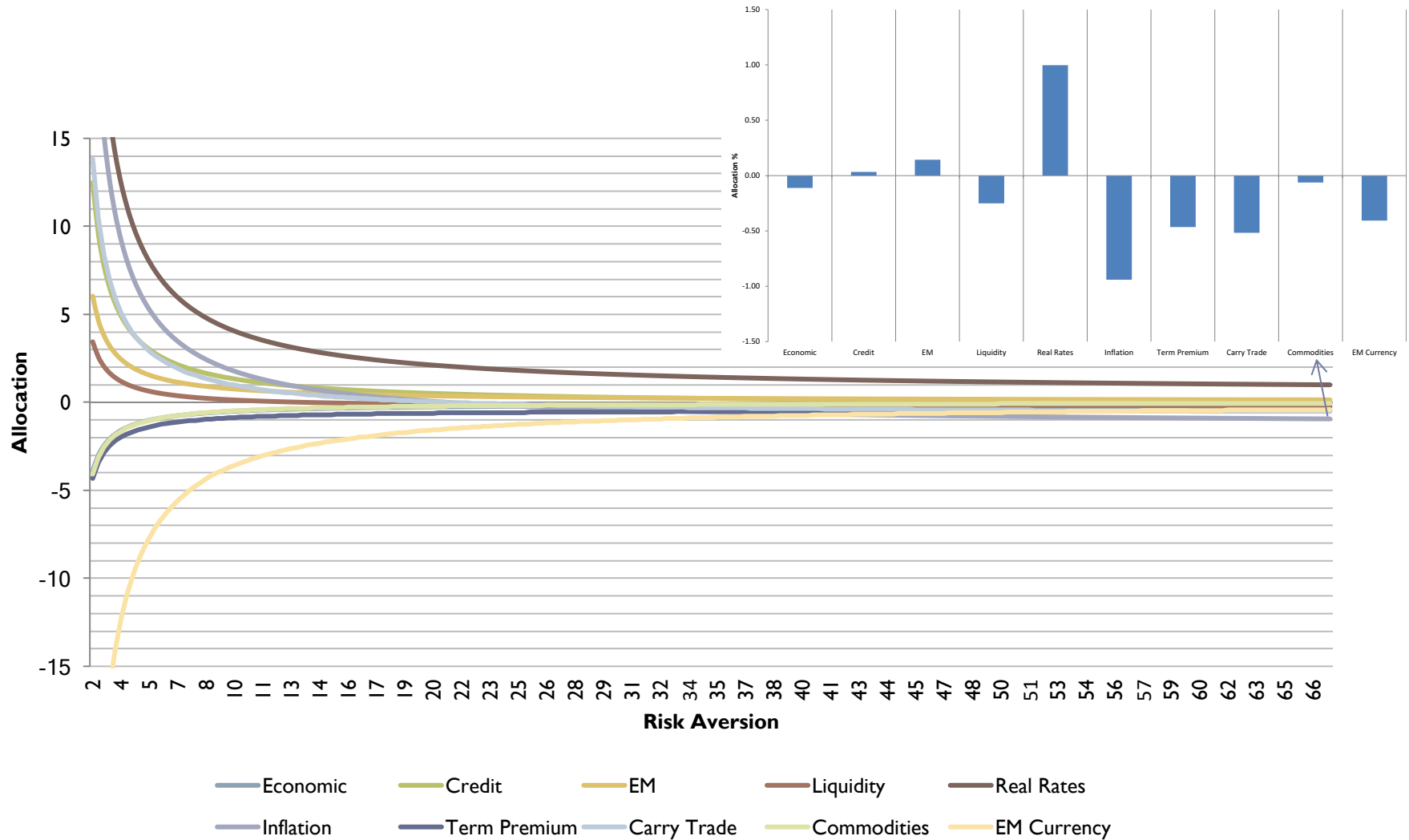
Anexos

- ▶ Eckhold (2010). The currency denomination of New Zealand's unhedged foreign reserves. Working Paper, New Zealand Central Bank.

The Reserve Bank's FX benchmark allocation

Currency	Benchmark allocation (%)	Tactical deviation range (%)
USD	25	+/- 5
EUR	25	+/- 5
JPY	5	+/- 5
GBP	15	+/- 5
CAD	10	+/- 5
AUD	20	+/- 5
Total	100	VaR limit NZD 5 m ⁶

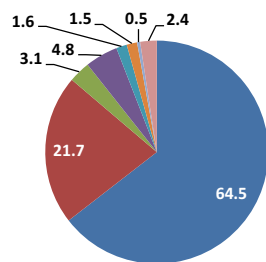
Optimal factor allocation



Diversification of foreign reserves

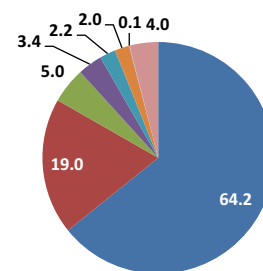
- ▶ During Bretton Woods (BW), Central Banks invested mainly in gold.

Reserves DM (2015)



■ USD ■ EUR ■ GBP ■ JPY ■ CAN ■ AUD ■ CHF ■ Others

Reserves EM (2015)



■ USD ■ EUR ■ GBP ■ JPY ■ CAN ■ AUD ■ CHF ■ Others

*Rodrik (2006) shows that the cost of holding reserves for EM countries can be easily account for 1% GDP.

Diversification of foreign reserves

- ▶ During Bretton Woods (BW), Central Banks invested mainly in gold.
- ▶ After the Asian Crisis, the literature started analyzing the trade-off between the ability to hedge sudden-stop risk, and the yield-give up cost.*
- ▶ Post-BW: Countries started diversifying away from the US dollar. Starting from European currencies (in the pre-Euro era), Great Britain Pound, Japanese Yen and the Swiss franc. Lately, countries have started including the Korean Won, Brazilian Real, Polish Złoty and Renminbi.
- ▶ Even more, some countries such as China, Japan, Korea, Switzerland (15%), Denmark, Italy, Israel (10%) and the Czech Republic have included stocks in their reserves' portfolio.



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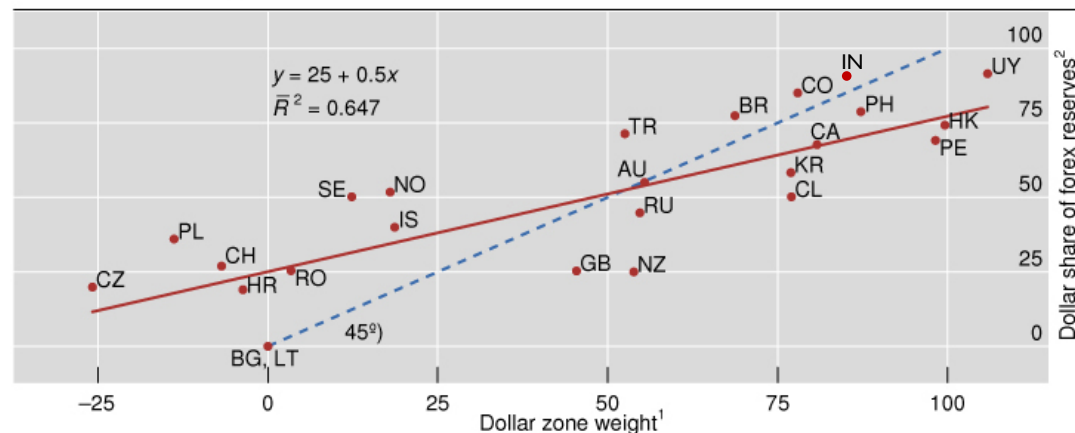


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Diversification of foreign reserves

- ▶ BIS (2014) indentifies “pseudo monetary zones”: USD, Yen y EUR.

Dollar zone weight and dollar share of forex reserves, 2013 Graph 3



AU = Australia; BG = Bulgaria; BR = Brazil; CA = Canada; CH = Switzerland; CL = Chile; CO = Colombia; CZ = Czech Republic; GB = United Kingdom; HK = Hong Kong SAR; HR = Croatia; IS = Iceland; KR = Korea; LT = Lithuania; NO = Norway; NZ = New Zealand; PE = Peru; PH = Philippines; PL = Poland; RO = Romania; RU = Russia; SE = Sweden; TR = Turkey; UY = Uruguay.

¹ Average over four years. ² For Australia, June 2013; Colombia, March 2011; Brazil, Croatia and Turkey, 2012; New Zealand, September 2010.

Sources: National data; BIS calculations.

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*The dollar share is calculated in two steps. First, for a given currency, its weekly percentage change against the dollar is regressed on the weekly percentage change of the euro/dollar and yen/dollar rates. The dollar zone weight is calculated as 1 minus the corresponding regression coefficients.

Methodology: Systematic risk of CB Liabilities

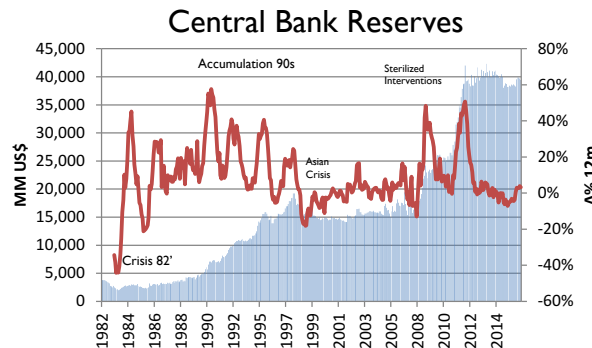
- ▶ Given the estimated historical returns of the portfolio of CB's liabilities, we can decompose the variance into systematic and idiosyncratic components:

$$r_t^n = c^n + B^n \tilde{f}_t + \tilde{\varepsilon}_t^n$$

where, c is constant, $\tilde{f}$ are systematic factors, B^n is a vector of factor loadings and $\tilde{\varepsilon}$ is the unexplainable component.

The case of Chile

- ▶ During the 80s, foreign Exchange reserves were used intensively in order to face the balance of payment crisis that Latin American countries faced as consequence of currency pegs and the monetary policy tightening in the US.
- ▶ During the 90s the CB accumulated reserves in order to maintain the exchange rate between a floating band.
- ▶ After the Asian Crisis the CB tried to defend the peso. The CB reacted raising rates from 9% to 19%, and selling US\$ 4 bn. The unsuccessful efforts ended up with a liberalization of the Exchange rate in September 1999.
- ▶ Since the Asian crisis, we have seen 4 sterilized interventions that have intended to tackle possible short-term misalignments.
- ▶ During the 2008 financial crisis played a different role. The CB offered repos/swaps in the local financial market as a response of the liquidity shock.

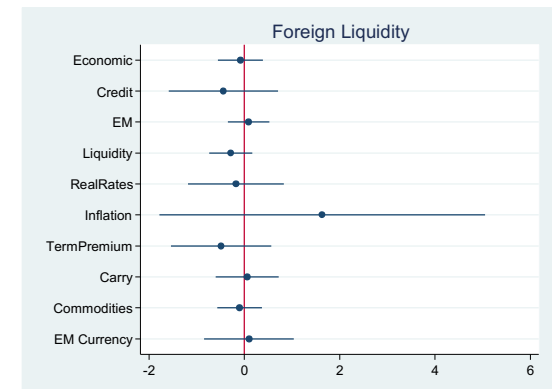
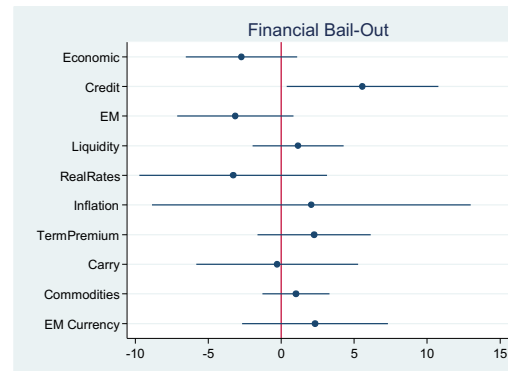
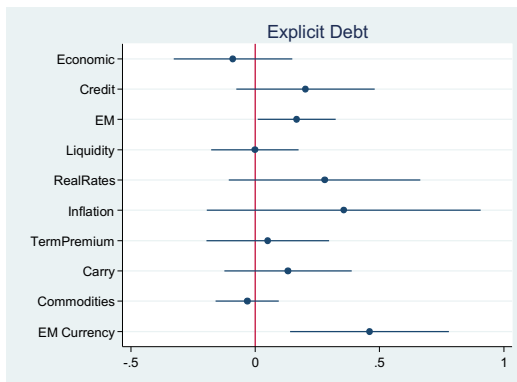
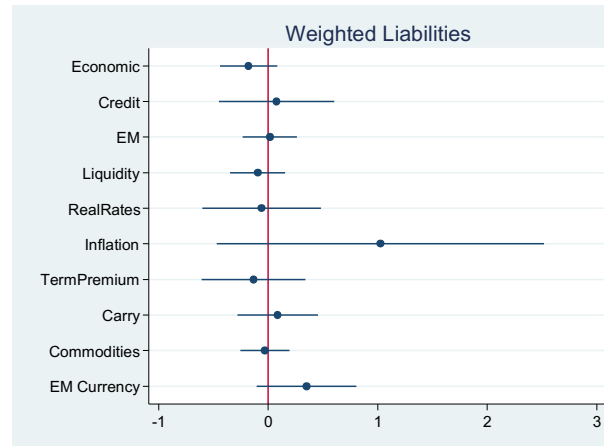


Market value of liabilities

- ▶ Explicit debt: Bond index of Government issued securities (49%).
- ▶ Contingent financial bail-out: Ronn & Verma (1986) methodology on publicly-traded Banks (3%).
- ▶ Foreign liquidity guarantee: Swaption valuation under Vasicek model on the IMF's foreign currency credit spread (48%).

Systematic and idysincratic risk decomposition of liabilities

- ▶ Time series regression of liabilities' returns on factors:



Risk premium parameters

<i>Panel A: US\$ Dollar Returns</i>	Period	Mean	Median	Std. Deviation
Economic	Feb-90/Jun-16	0.04%	0.35%	3.09%
Credit	Feb-90/Jun-16	0.25%	0.31%	2.51%
Emerging Markets (EM)	Feb-90/Jun-16	0.20%	0.20%	4.35%
Liquidity	Feb-95/Jun-16	0.20%	0.31%	2.39%
Real Rates	Jan-97/Jun-16	0.51%	0.62%	2.14%
Inflation	Jan-97/Jun-16	-0.09%	-0.13%	1.37%
Term Premium	Feb-90/Jun-16	0.47%	0.48%	2.73%
Carry Trade	Feb-90/Jun-16	0.28%	0.48%	2.60%
Commodities	Feb-91/Jun-16	-0.17%	-0.09%	4.27%
Emerging Markets Currencies (EMC)	Nov-96/Jun-16	-0.98%	-0.43%	3.32%

Economic	0.648	0.264	0.213	0.417	-0.406	-0.118	0.570	0.340	0.618
0.648	Credit	0.393	0.330	0.262	-0.522	-0.358	0.523	0.395	0.553
0.264	0.393	EM	0.361	0.306	-0.285	-0.037	0.382	0.391	0.542
0.213	0.330	0.361	Liquidity	0.098	-0.182	-0.068	0.185	0.228	0.182
0.417	0.262	0.306	0.098	RealRates	-0.494	0.413	0.426	0.486	0.502
-0.406	-0.522	-0.285	-0.182	-0.494	Inflation	0.090	-0.619	-0.375	-0.356
-0.118	-0.358	-0.037	-0.068	0.413	0.090	TermPremium	-0.148	-0.150	-0.171
0.570	0.523	0.382	0.185	0.426	-0.619	-0.148	Carry	0.480	0.615
0.340	0.395	0.391	0.228	0.486	-0.375	-0.150	0.480	Commodities	0.540
0.618	0.553	0.542	0.182	0.502	-0.356	-0.171	0.615	0.540	EM Currency